CarePulse | Interactive Plotly Visualization

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Synthetic diabetes-model cohort • Reproducible seed: 42 • 2,500 records

Data note: figures use the reproducible synthetic cohort generator already present in CarePulse. They do not contain patient records, validate a clinical model, or support individual medical decisions. Simulator outputs are labeled as such.

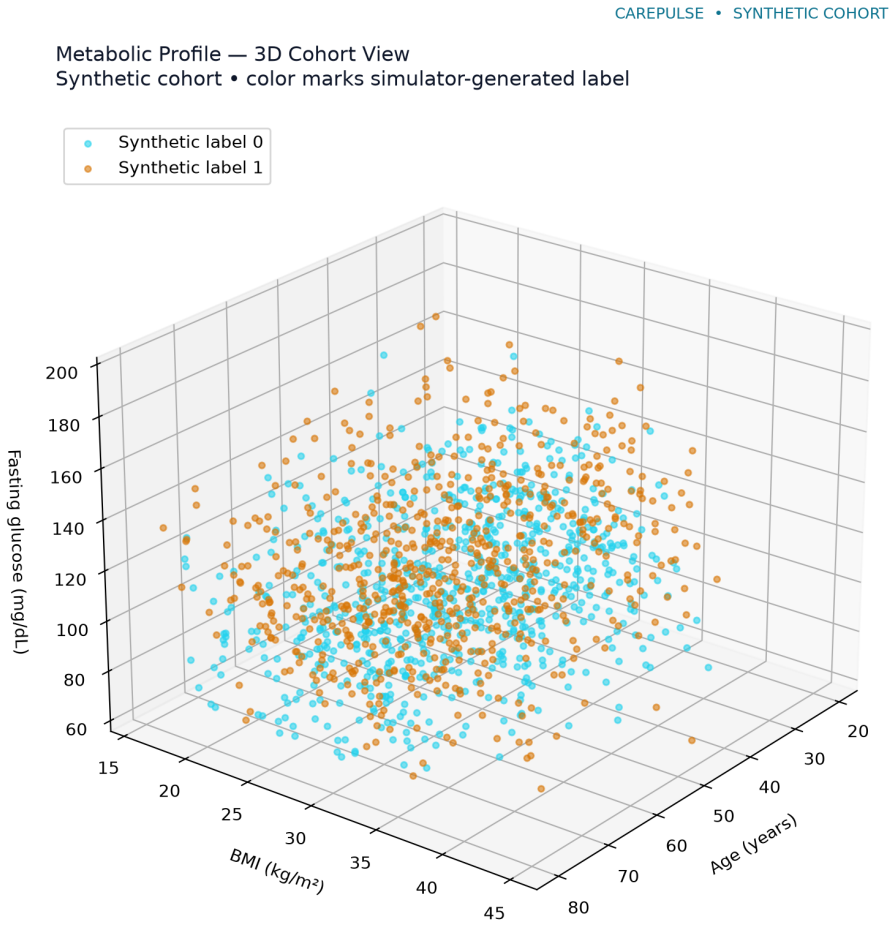
Static submission: PNG and PDF figures do not support hover, orbit, zoom, or brushing. Those interactive controls are retained in the Plotly figure definitions in the Python source.

Generated cohort label prevalence: 45.6%. This is a property of this simulated sample only.

Metabolic Profile — 3D Cohort View

Each point is a synthetic cohort record. Age, BMI and fasting glucose form the 3D axes; color distinguishes the generator's binary outcome label. Hover reveals additional features and the generating probability. This descriptive view does not establish clinical risk for real people.

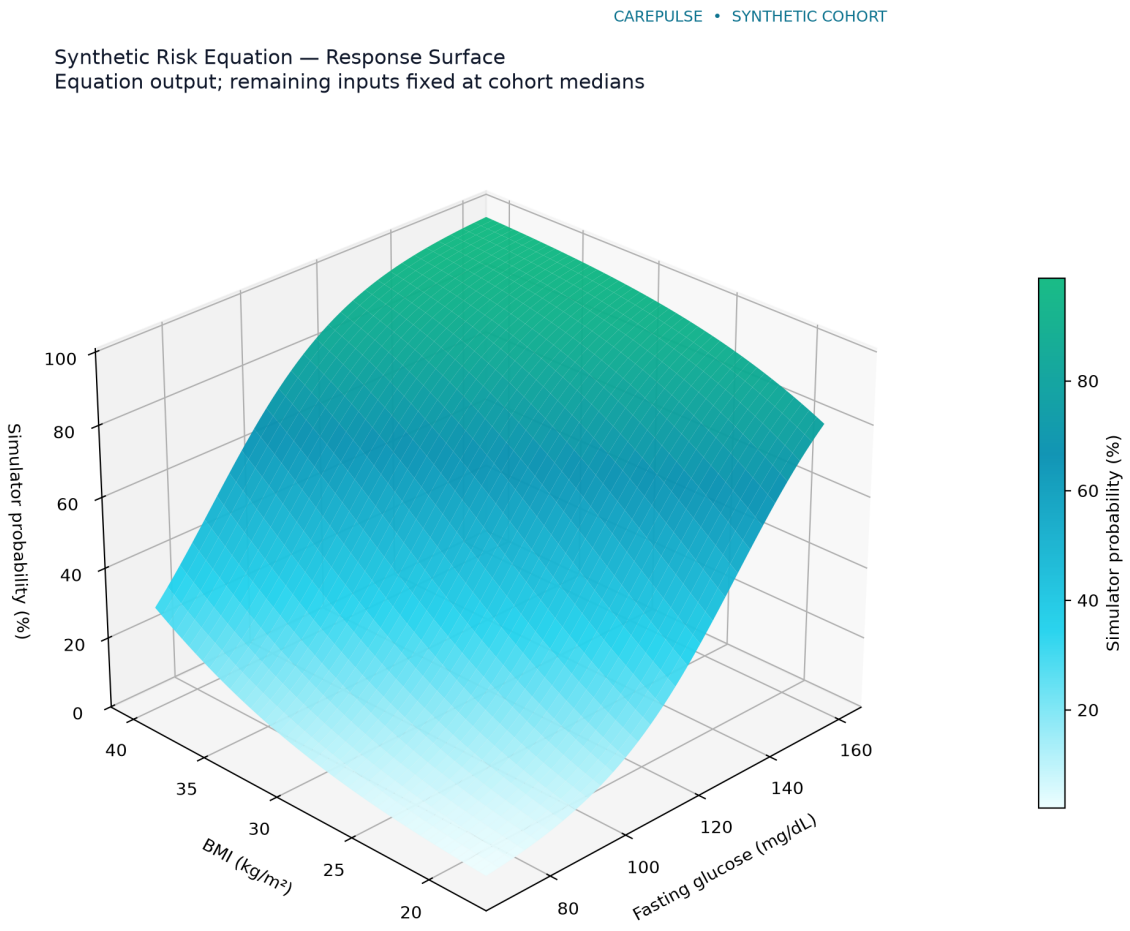
Variables: Age (years), BMI (kg/m²), fasting glucose (mg/dL); color: simulator-generated outcome.



Synthetic Risk Equation — Response Surface

The surface evaluates the logistic probability equation already used to generate the project's diabetes training cohort. Glucose and BMI vary; age, systolic pressure, activity and family history are held at synthetic-cohort medians. It illustrates the simulator only and is not a clinical prediction model.

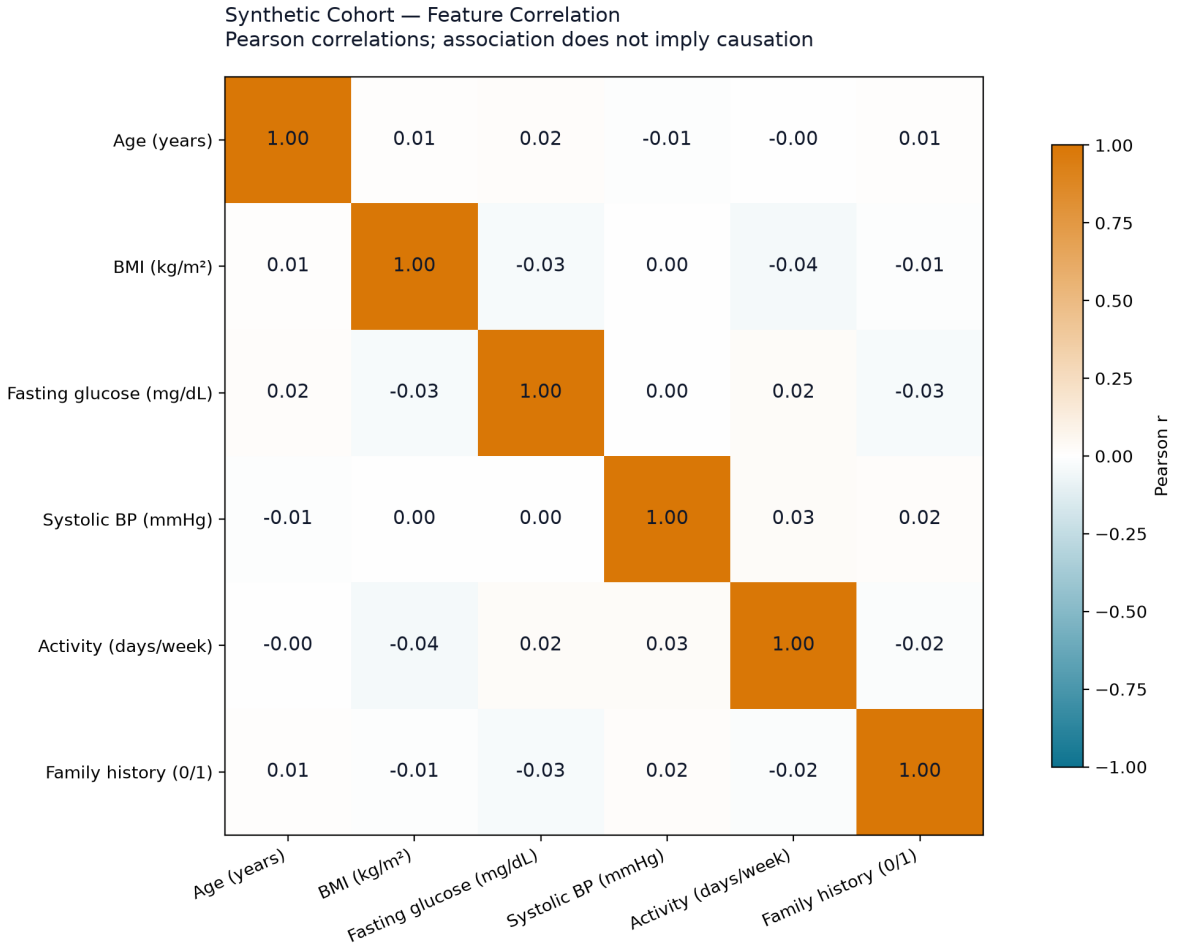
Variables: Fasting glucose (mg/dL), BMI (kg/m²); height and color: generator probability (%).



Synthetic Cohort — Feature Correlation

This Pearson correlation matrix summarizes pairwise linear association among the six diabetes-model inputs. Values near zero indicate weaker linear association within this generated sample; the matrix is descriptive and does not imply cause or effect.

Variables: Age, BMI, fasting glucose, systolic BP, activity days/week and family history.



Multivariable Metabolic Profiles

Parallel coordinates show the distribution of six input dimensions across a sample of synthetic records. Each line is one record; color maps to the data generator's probability. The plot supports interactive brushing in Plotly's source figure.

Variables: Age, BMI, fasting glucose, systolic BP, activity, family history; color: simulator probability.

